



Normal Urine

College of Medicine - HMU

1st year Medical Students

Prepared by

Shelan I. Saber

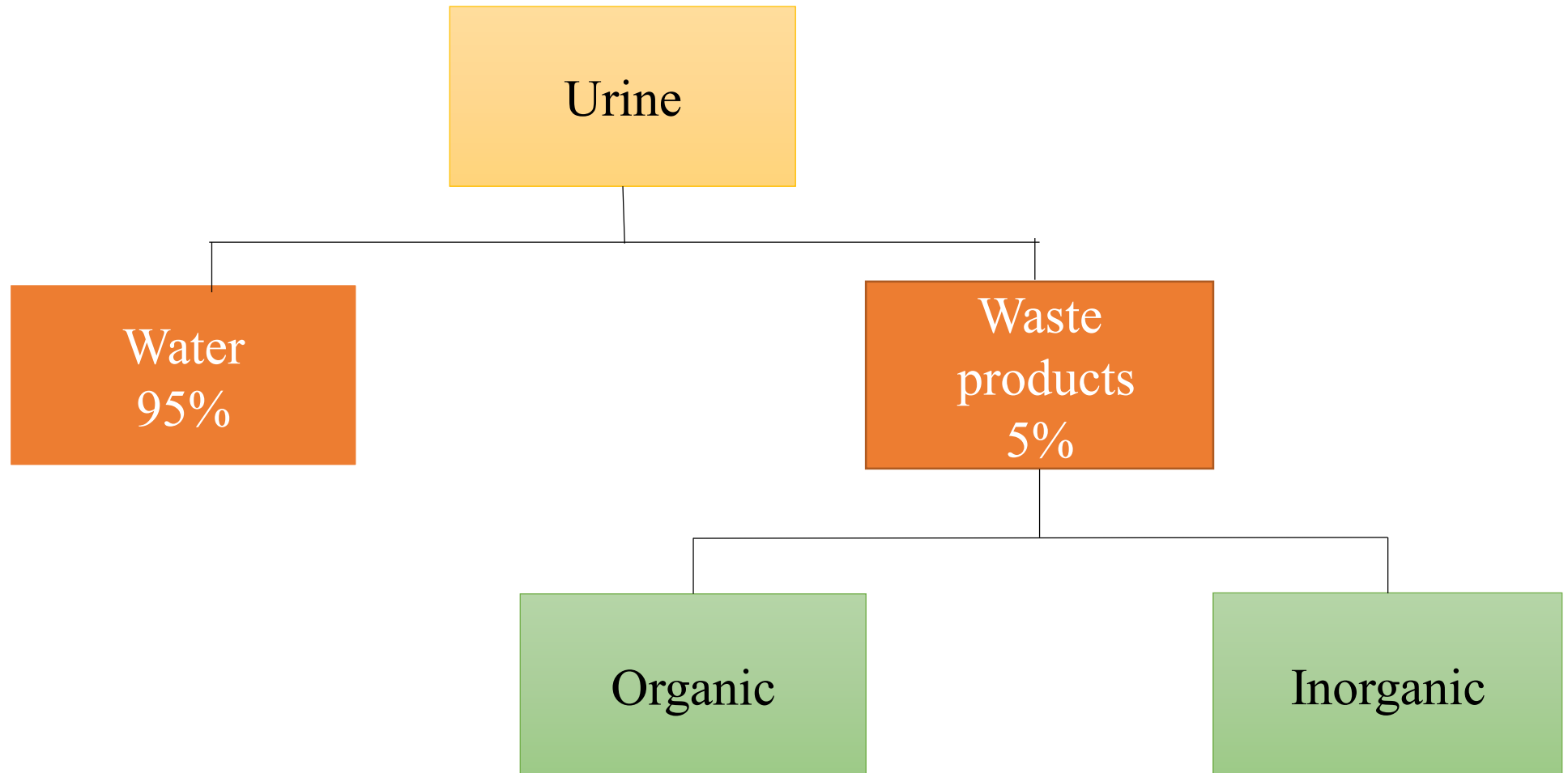
MSc. in Biochemistry

Urine

Urine is composed of water, certain electrolytes and various waste products that are filtered out of the blood system.

Human urine is composed primarily of water (95%). The rest being made up of organic nitrogenous compounds is urea (2%), creatinine (0.1%), uric acid (0.03%) and inorganic salts like chloride, sodium, potassium, sulphate, ammonium, phosphate and other ions and molecules in lesser amounts.

Composition of normal urine

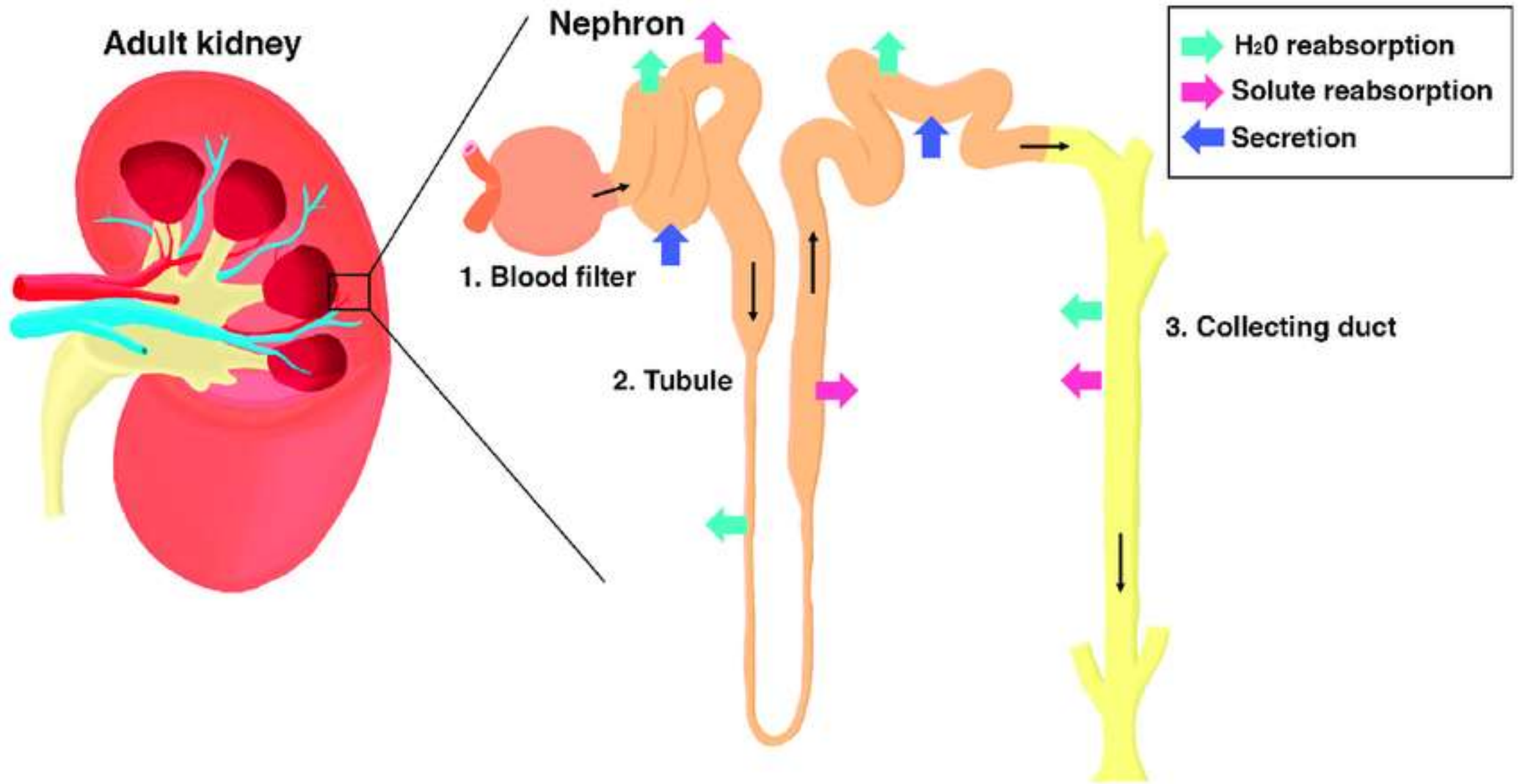


Normal daily excretion of urinary solutes

<i>Compound</i>	<i>Total daily excretion under normal dietary intake</i>	
Sodium	100 – 200 mmol	(2–4 g)
Potassium	50 – 70 mmol	(1.5–2 g)
Magnesium	4 – 8 mmol	(0.1–0.2 g)
Calcium	1.2 – 3.7 mmol	(0.1–0.3 g)
Phosphate	20 – 50 mmol	(0.7–1.6 g)
Chloride	100 – 250 mmol	
Bicarbonate	0–50 mmol	
Sulfate (inorganic)	40 – 120 mEq	(0.6–1.8 g)
Sulfate (organic)	—	(0.06–0.2 g)
Urea	15 – 30 g	(6–18 gN)
Creatinine	1–2 g	(0.3–0.8 gN)
Uric acid	0.5–0.8 g	(0.08–0.2 gN)
Ammonia	30–75 mEq	(0.04–1 gN)
Amino acids		(0.08–0.15 gN)

The function unit of kidney is nephron which has two main parts:-
glomerular and tubular which forms urine by three regulated
processes :-

1. Glomerular filtration of water and solutes from the blood.
2. Tubular reabsorption of water and conserved molecules back into the blood.
3. Tubular excretion (secretion) of ions and other waste products from surrounding capillaries.



Kidneys perform a number of functions:

1. Maintain volume of extracellular fluid
2. Maintain ionic balance in extracellular fluid
3. Maintain the pH (acid-base balance) concentration of the extracellular fluid.
4. Excrete toxic metabolic by-products such as urea, ammonia, and uric acid.
5. Endocrine functions

The human kidney secretes three hormones

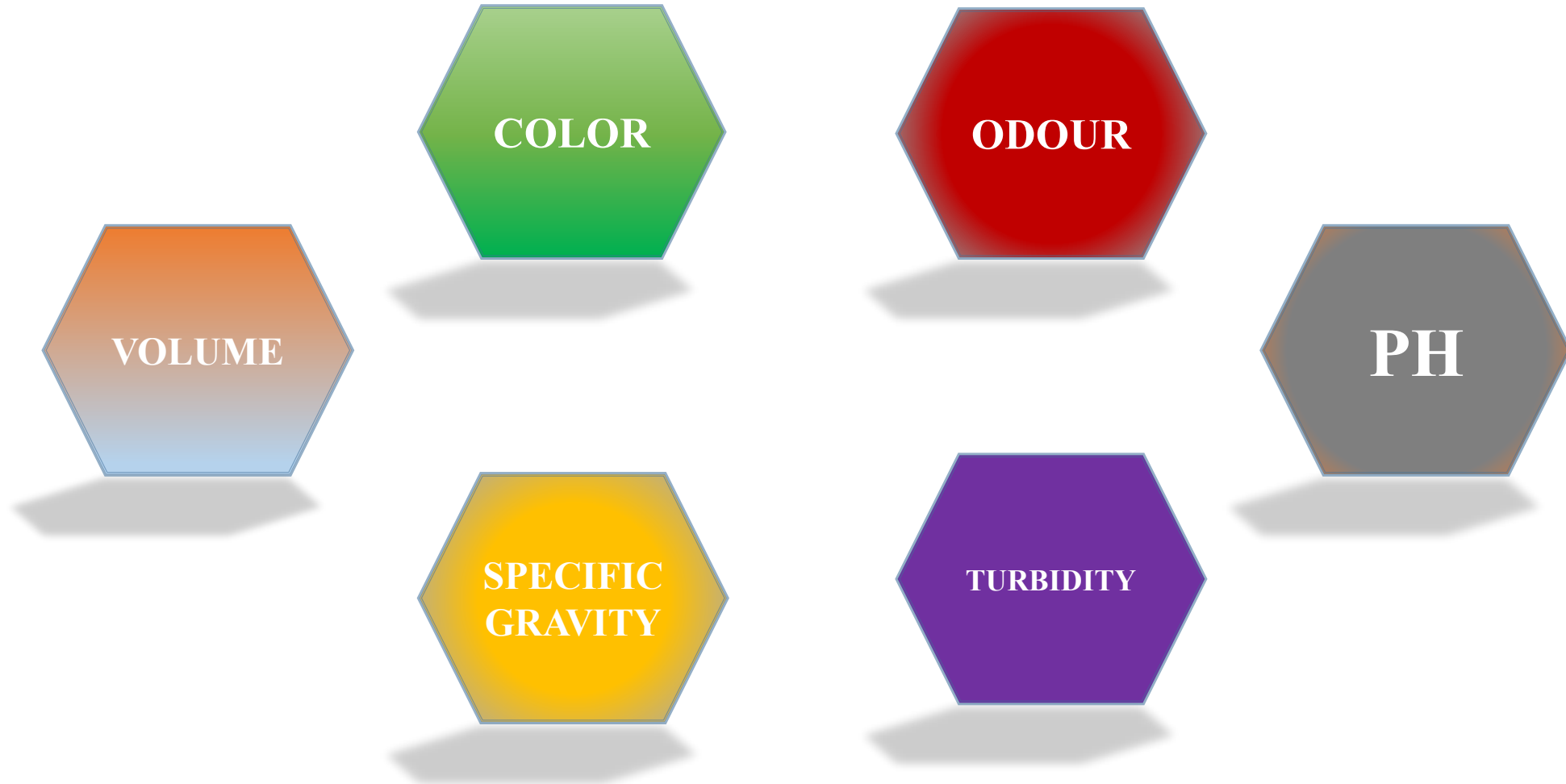
1- Rennin

2-Erythropoietin (EPO)

3- Calcitriol [1,25-dihydroxy cholecalciferol]

active vitamin D3

Physical properties of urine



Physical properties of urine

1. Volume:-

A normal adult excretes 800 to 2000 ml of urine per day. The amount depends on the liquid intake, the surrounding temperature, and diet. Drugs such as caffeine have a diuretic effect, meaning they increase the flow of the urine. A decreased flow of urine is called oliguria (Certain kidney diseases (acute nephritis)) cause oliguria.

2. Specific gravity (density):-

The specific gravity of urine depends upon the concentration of the solute.

The greater concentration of solute in urine means the greater specific gravity.

The normal range of specific gravity of urine is (1.003 – 1.030).

$$\text{Sp.gr} = \text{wt. of urine in certain volume} / \text{wt. of water in the same volume}$$

3. Color:-

Normal urine ordinarily possess tint yellow tend to red. Under the pathological conditions or after using of various drugs the color of urine may vary in intensity from light yellow to very dark brown or may even assume the color of the drugs or their degradation products.

The color of urine may change in large number of pathological cases such as:-

1. Liver diseases:- the color becomes deep yellow to brown.
2. Bile pigment:- the color becomes green.
3. Blood:- the color becomes red.

4. Turbidity (transparency):-

Normal urine is ordinarily perfectly clear and transparent when voided.

Turbidity due to the precipitation of phosphates is normally noted in urine passed after a heavy meal.

Turbid urine ordinarily arise from pathological conditions.

5. pH (acidity – alkalinity) :-

Urine is normally slightly acidic with a pH range of (4.6 – 8.0) and an average value about 6.3, however the pH of urine varies with the diet.

6. Odor (smell):-

The odor of urine is a faint aromatic type due to the presence of small amount of certain volatile organic acids. When urine allowed standing for a long time it undergoes decomposition and very unpleasant odor is evolved.

Normal constituents of urine

Average samples of urine contain about 4% solids dissolved in it, approximately 2% is urea, 1% is sodium chloride and all other constituents make up the remaining 1%.

Urea, uric acid and creatinine are the principle nitrogenous compounds of normal urine. Among the inorganic substances present in urine are chloride, phosphate and sulphates of sodium, calcium and magnesium.

1. Inorganic substances:

A. Chloride:-

The average adult excretion of sodium chloride is approximately 10gm in 24 hours.

Principle:-

Silver nitrate in concentrated nitric acid is added to urine forming a white precipitate of silver chloride



B. Sulphate (SO_4^{-2}):-

Principle:-

Barium chloride in concentrated HCl is added to urine, a milky solution of barium sulphate indicate the presence of inorganic sulphate in urine.

C. Phosphate (PO_4^{-3}):-

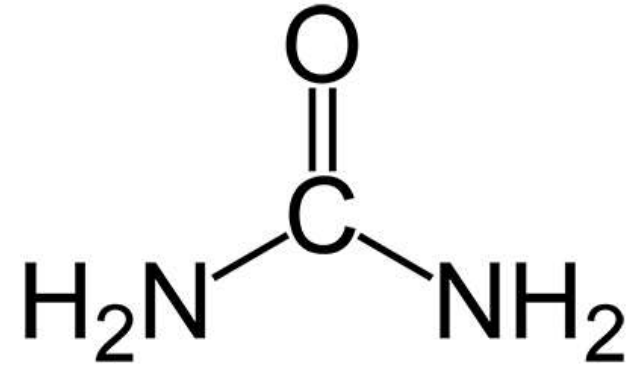
Principle:-

Phosphates are present in various forms in urine, inorganic phosphate (phosphates of K^+ , Na^+ and NH_4^+) and organic phosphate (phospholipids, phosphoproteins and nucleoproteins and others).

Pathologically the excretion of phosphate increased in the bone diseases as osteomalacia and rickets.

2. Organic compounds:-

A. Urea:-



Urea is one of the main organic nitrogenous constituent of urine. About (80-90%) of nitrogen in the urine is in the form of urea.

B. Uric acid:-

A product of the metabolism of purine from nucleoprotein, it is only slightly soluble in water and excreted primarily as urate salt.

The average daily excretion of uric acid is about 0.7gm, but an increase in nucleoproteins in the diet will cause an increased excretion of uric acid, the output of uric acid is also increased in leukemia and severe liver disease.

URIC ACID STRUCTURAL FORMULA

BYJU'S
The Learning App



C. Creatinine:-

Creatinine is the end product of protein metabolism. The average adult daily excretion is 1.4gm for creatinine.

